

Unit 1 Mini Project

Enterprise Student Record and Academic

Design Thinking & Product Planning Report

Prepared as part of the DevOps / Product Engineering Training Program

Domain: Higher Education — Academic & Student Record Management

Name : Anuraj Balasaheb Tambe

PRN : 2125UDSM1043

Dept : AI-DS

Table of Contents

1. Objective & Business Understanding Document (BRD)
2. Design Thinking Report
3. IBM Design Thinking Report & IBM Loop Documentation
4. Stakeholder Analysis Report
5. User Persona Report
6. Functional Requirements Specification (FRS)
7. Non-Functional Requirements Specification (NFRS)
8. Product Vision Document
9. Minimum Viable Product (MVP) Definition
10. Product Roadmap
11. Initial System Architecture Diagram
12. Project Charter
13. Project Repository Initialization & Folder Structure
14. Product Discovery Documentation & Deliverables Checklist

1. Objective & Business Understanding Document (BRD)

1.1 Objective

The objective of this project is to apply Design Thinking to the planning of a software platform for managing student records and academic information in a higher-education environment. The work begins with understanding the academic domain, identifying stakeholders, studying their needs, defining requirements, and establishing a practical product vision and roadmap before Agile development begins.

1.2 Business Domain Overview

Higher-education institutions handle student information throughout a continuous lifecycle that includes admission, enrollment, course registration, attendance, examinations, and results. When these activities depend on paper records or disconnected spreadsheets, information becomes difficult to maintain, update, and share accurately.

Core Academic Operations

- Admissions & Enrollment: Capture applicant information, verify eligibility, and complete enrollment into programs and academic years.
- Course Registration: Enable students to select semester courses and electives while following applicable rules.
- Attendance Management: Record attendance by class or session and determine eligibility against required thresholds.
- Examination Management: Schedule examinations, record internal and external marks, and calculate grades.
- Result & Report Management: Produce mark sheets, transcripts, and academic progress reports for students and authorized staff.

1.3 Problems with Manual Record Management

Manual records can be distributed across departments, creating duplication and inconsistencies. A centralized relational database can provide a single dependable source of information and target approximately 99.9% record accuracy with minimal duplicate records.

Paper attendance registers require repeated manual processing. Digital attendance capture and automatic aggregation can reduce attendance-processing effort by more than 80%.

Manual examination calculations and mark-sheet preparation can delay publication for weeks. Automated result computation and transcript generation can shorten the turnaround to hours.

Without live dashboards, administrators often need days to compile reports. Role-based dashboards and analytics can reduce reporting work from days to minutes.

Disconnected communication among admissions, faculty, and examination staff can introduce data errors. A shared platform with controlled role-based access can substantially reduce cross-department inconsistencies.

1.4 Business Justification

Bringing student records, attendance, and examination results into one platform can lower administrative effort, improve information accuracy, and support faster academic decisions. Students also benefit through self-service access to their own academic information rather than depending on office visits.

2. Design Thinking Report

Overview

Design Thinking is a user-centered, iterative approach that emphasizes understanding people and their problems before finalizing technical solutions. The five stages below are applied to the Student Record and Academic Management Platform.

Stage 1: Empathize

The principal user groups considered in their academic workflows are Students, Faculty Members, Academic Administrators, Examination Cell Staff, and IT Support Personnel. Each group faces different difficulties when records are maintained manually or across disconnected systems.

Stage 2: Define

Core Problem Statement Students and academic staff face delays and avoidable errors while registering for courses, recording attendance, and obtaining examination results because records are fragmented and often maintained manually. Administrators also lack a centralized real-time view, which contributes to reporting delays, inconsistent information, and lower institutional efficiency.

Stage 3: Ideate

Potential high-value features identified during brainstorming include: • A self-service student portal for registration, attendance, and results. • Automatic attendance eligibility calculations and shortage notifications. • A real-time engine for result and grade calculation. • Role-based administrative dashboards with downloadable reports. • Notifications for examination schedules, deadlines, and important announcements.

Stage 4: Prototype

Initial low-fidelity concepts were planned for the main interface modules: Student Login & Dashboard, Course Registration Screen, Faculty Attendance & Marks Entry Console, and the Admin Analytics Panel.

Stage 5: Test

The proposed flows were checked against representative student, faculty, and administrator situations to assess navigation, ease of data entry, and access to reports before backend implementation.

3. IBM Design Thinking Report & IBM Loop Documentation

3.1 IBM Design Thinking Pillars

Focus on User Outcomes: Instead of simply building an attendance table, the intended outcome is to let a student check attendance eligibility quickly, ideally within a few interactions.

Restless Reinvention: Continuous deployment is treated as an ongoing improvement cycle in which usage information is reviewed and used to refine registration workflows and dashboard performance.

Diverse Empowered Teams: The platform benefits from cooperation among Product Owners, UX Designers, Software Engineers, QA Engineers, Database Administrators, and DevOps Engineers.

3.2 The IBM Loop: Observe → Reflect → Make

The loop is applied repeatedly during development sprints. Usage information from registration and attendance features is observed, reviewed against institutional performance indicators, and converted into improvements, refinements, or fixes.

4. Stakeholder Analysis Report

4.1 Identified Stakeholders

- Students — primary users who register for courses and access academic records.
- Faculty Members — maintain attendance, enter marks, and communicate academic updates.
- Academic Administrators — supervise academic operations and compliance.
- Examination Cell Staff — handle examination schedules, results, and transcripts.
- IT Support Personnel — maintain availability, security, and technical support.

4.2 Stakeholder Responsibility Matrix

Business Stakeholders — College Management, Academic Administrators, Students, and Faculty. They define the business direction, validate usability, and ensure academic-policy compliance.

Engineering Stakeholders — Software Developers, UI/UX Designers, and QA Engineers. They design the codebase, create responsive interfaces, perform automated testing, and maintain quality.

Operations Stakeholders — DevOps Engineers, Database Administrators, and the Security Team. They provision infrastructure, protect database integrity, automate CI/CD, and maintain monitoring.

Examination Cell — Exam Controller and Result Processing Staff. They validate examination workflows, result-calculation rules, and transcript formats.

5. User Persona Report

Overview

The personas below describe representative users, their objectives, everyday responsibilities, frustrations, and the platform capabilities most relevant to them.

Ananya — Student

Goals: Complete course registration on time and monitor attendance and results without repeatedly visiting the office. Frustrations: Registration queues, late result announcements, and limited visibility of attendance. Key Feature: Self-service access to course registration, attendance tracking, and digital mark sheets.

Prof. Mehta — Faculty

Goals: Enter attendance and marks efficiently and share updates with students. Frustrations: Paper attendance, repeated data entry, and pressure around delayed grade submissions. Key Feature: Digital attendance and marks-entry console with validation and bulk-upload support.

Dr. Kapoor — Academic Administrator

Goals: Track institutional academic performance and prepare compliance or accreditation reports. Frustrations: Lack of real-time dashboards and time-consuming compilation of information from multiple departments. Key Feature: Executive analytics dashboard with quick report export.

6. Functional Requirements Specification (FRS)

Functional Requirements

- FR-01 Authentication: Provide secure role-based access for Admin, Faculty, Student, and Examination Cell users.
- FR-02 Student Registration: Support creation, viewing, updating, and deletion of student profiles, program enrollment, and academic-year assignments.
- FR-03 Course Registration Engine: Provide semester-wise course selection with prerequisite and seat-capacity validation.
- FR-04 Attendance Management: Capture attendance by session, aggregate it, and calculate eligibility thresholds.
- FR-05 Examination & Result Management: Support marks entry, automatic grade calculation, and transcript generation.
- FR-06 Reporting Engine: Generate configurable academic and administrative reports that can be exported to PDF or Excel.
- FR-07 Notification System: Send automated alerts for registration deadlines, attendance shortages, and result publication.

7. Non-Functional Requirements Specification (NFRS)

Non-Functional Requirements

- NFR-01 Security: Use secure password hashing such as Bcrypt/Argon2, JWT-based sessions, and HTTPS for network traffic.
- NFR-02 Performance: Target API responses below 500 ms for search and dashboard queries under concurrent load.
- NFR-03 Reliability & Availability: Use containerized deployment with a target of 99.9% availability and automated failover.
- NFR-04 Scalability: Allow horizontal scaling so the system can support institution-wide concurrent use during registration and result periods.
- NFR-05 Usability: Provide a responsive and accessible interface across desktop and mobile devices with minimal training.
- NFR-06 Maintainability: Keep the code modular and use Infrastructure as Code (IaC) for consistent, repeatable deployment.

8. Product Vision Document

8.1 Vision Statement

To provide a centralized, secure, and easy-to-use Student Record and Academic Management Platform that reduces manual administration, simplifies registration and attendance activities, and gives students, faculty, and administrators timely access to reliable academic information.

8.2 Mission Statement

The mission is to support higher-education institutions with a dependable and scalable digital platform that replaces fragmented manual procedures with one trusted source of academic information.

8.3 Success Criteria

- Approximately 99.9% accuracy in student records with no unrecorded duplicates.
- More than 80% reduction in attendance and result-processing time.
- Core search and dashboard operations responding in under 500 ms.
- Positive adoption feedback from students, faculty, and administrators during the first semester.

8.4 Business Objectives

- Reduce administrative work caused by manual record maintenance.
- Increase transparency and self-service access for students and faculty.
- Give administrators real-time, data-driven academic insights.
- Build a scalable technical base that can later accommodate modules such as fees, hostel, and library.

9. Minimum Viable Product (MVP) Definition

MVP Scope

To keep the first release focused and manageable, non-essential capabilities such as AI-based academic recommendations and third-party payment integration are postponed. Sprint 1–4 concentrates on the core academic functions:

- Student and Faculty login with role-based access control.
- Student registration and profile management.
- Course registration.
- Attendance capture and basic eligibility monitoring.
- Examination result entry and mark-sheet generation.
- Basic administrative reporting dashboard.

10. Product Roadmap

Roadmap Approach

The roadmap follows the progression of the training program. Unit 1 establishes the business understanding and design foundations, covering the initial planning phases. Later units are intended to address Agile planning, software development, and DevOps automation.

11. Initial System Architecture Diagram

Layered Module View

The proposed platform uses a layered architecture so that user interfaces, business logic, data storage, and external integrations remain separated. This separation supports maintainability, controlled access, and future expansion of the system.

12. Project Charter

Project Charter

Project Name: Enterprise Student Record and Academic Management Platform

Project Purpose: Replace manual academic record-keeping with a centralized digital platform.

Sponsor: College Management / Academic Administration.

Project Lead: Product Owner (Training Cohort).

Scope (Unit 1): Business understanding, design thinking, requirements, product vision, MVP definition, and roadmap planning.

Out of Scope (MVP): AI-based recommendations, third-party payment gateways, and hostel/library integration.

Key Milestones: Domain analysis → Design thinking → Requirements → Vision & MVP → Repository setup.

Success Measures: 99.9% record accuracy, 80% faster attendance/result processing, and positive user adoption.

13. Project Repository Initialization & Folder Structure

Repository Setup

A Git repository is established for the platform using a standardized folder structure to support collaborative development in later units. The README.md file is maintained at the repository root alongside the project folders.

Git Initialization Steps

- git init — create and initialize the local repository.
- Create the required project folder structure.
- git add . && git commit -m "Initial project structure" — stage and record the initial files.
- Connect the local repository to a remote repository and push the initial commit.

14. Product Discovery Documentation & Deliverables Checklist

The following deliverables were produced during Unit 1 as part of the product discovery and planning process, in preparation for Agile sprint planning in subsequent units.

- **[Completed]** Business Requirements Document (BRD)
- **[Completed]** Product Vision Document
- **[Completed]** Stakeholder Analysis Report
- **[Completed]** User Persona Report
- **[Completed]** Functional Requirements Specification (FRS)
- **[Completed]** Non-Functional Requirements Specification (NFRS)
- **[Completed]** Design Thinking Report
- **[Completed]** IBM Design Thinking Report
- **[Completed]** IBM Loop Documentation
- **[Completed]** MVP Definition
- **[Completed]** Product Roadmap
- **[Completed]** Initial System Architecture Diagram
- **[Completed]** Project Charter
- **[Completed]** Project Repository Initialization
- **[Completed]** Project Folder Structure
- **[Completed]** Product Discovery Documentation

End of Unit 1 Mini Project Report — Enterprise Student Record and Academic Management Platform.